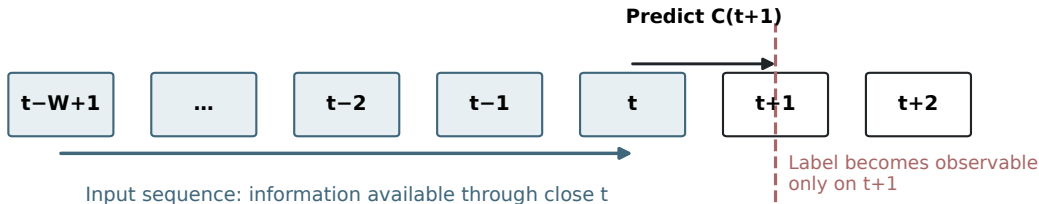


A Prediction-time contract at an evaluation boundary



Boundary rule: retain a training row only when Label_Date < first evaluation Date

B Expanding model selection and frozen outer evaluation

	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Selection 1							V							
Selection 2								V						
Selection 3									V					
Selection 4										V				
Outer 1											T			
Outer 2												T		
Outer 3													T	
Outer 4														T

Training-only fitting
Scaler · feature selection · model weights
regime thresholds · SHAP ranks

Frozen windows
LSTM 5 · CNN 20
LSTM-CNN 20 · LSTM-Attention 10
LSTM-CNN-Attention 20

Legend: Training

Validation

Outer test

Seeds: 42, 123, 456, 789, 2025